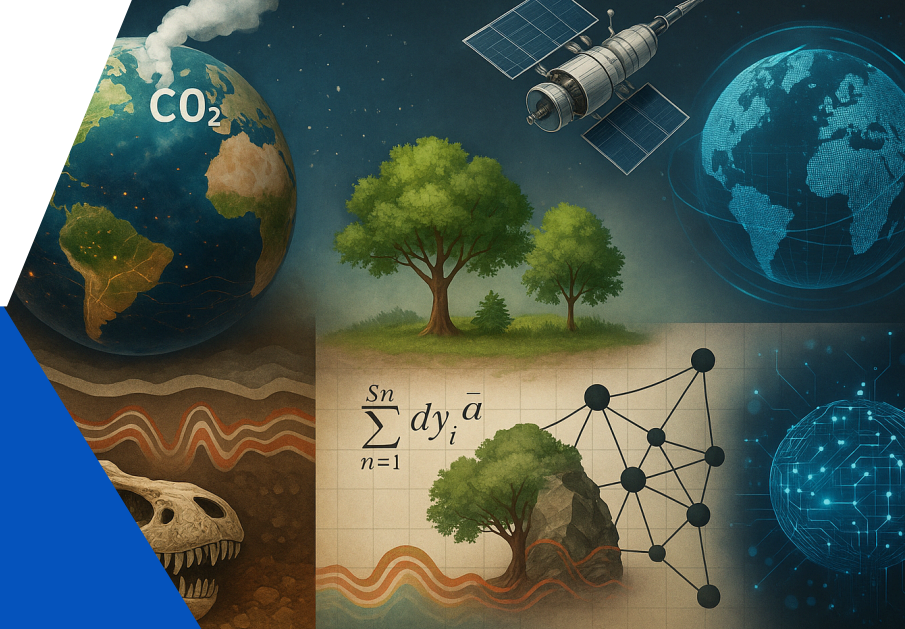




Deccan Education society's
Fergusson College
(Autonomous)

Interdisciplinary Workshop

Lessons and Challenges in Geophysical Data Analysis



23-27 June 2025

 Fergusson College, Pune

✓ Objective

To explore how mathematics, statistics, and machine learning can help solve real-world geology and geophysics problems, using interdisciplinary tools, data, and models.

✓ Core Themes

- Geophysical Data Analysis without error bars
- Applications of Machine Learning in Earth Sciences
- Interdisciplinary Research Opportunities
- Hands-on Data Projects and Tools like iLOSCAR

✓ TY/MSc:

- Geology | Economics | Mathematics | Physics | Statistics | Data Science (QGIS) | IMCA
- Limited to 40 students only*

✓ Pre-requisites

- Intro Statistics & Probability
- Linear Algebra
- Programming (Python preferred)
- Basics of Geology, Economics
- Differential Equations



**Reserve Your
Spot Now!**

(Limited slots available)

contact

1. Vrushali Limaye	-	9923396615
2. Lochana Mahajan	-	9922910559
3. Nishigandha Harapale	-	8975486231

✓ Abstracts

✓ Hamsa Padmanabhan

Lessons from Earth's Deadliest Extinctions

Analyzes the Permian Triassic Mass Extinction and its parallels with today's CO₂ crisis using data from isotope studies, climate models, and fossil evidence.

✓ Aboli Kulkarni

From Equations to Ecosystems

Explores how quantitative methods (stats, ML) help model biodiversity, carbon cycles, and ecological patterns—bridging field data with conservation insights.

✓ Girish Kulkarni

Earth, Data, and Algorithms

Presents ML techniques for complex geo datasets, addressing uncertainty, noise, and pattern recognition, encouraging interdisciplinary collaboration.

✓ Hrishikesh Khaladkar

Inverse Theory in Geophysics

Covers extracting hidden geophysical structures from observational data using techniques like seismic tomography and resistivity imaging.

✓ Anay Shende

Industrial & Regulatory Applications

Real-world case studies from Hawaii: resource management, disaster forecasting, and advanced geophysical tools like airborne and borehole surveys.

✓ Ishita Ghoshal

Geo-Economics and Strategic Resources

How economic tools are used in geopolitical strategies—through trade, subsidies, investment—with examples and data analysis using econometrics.

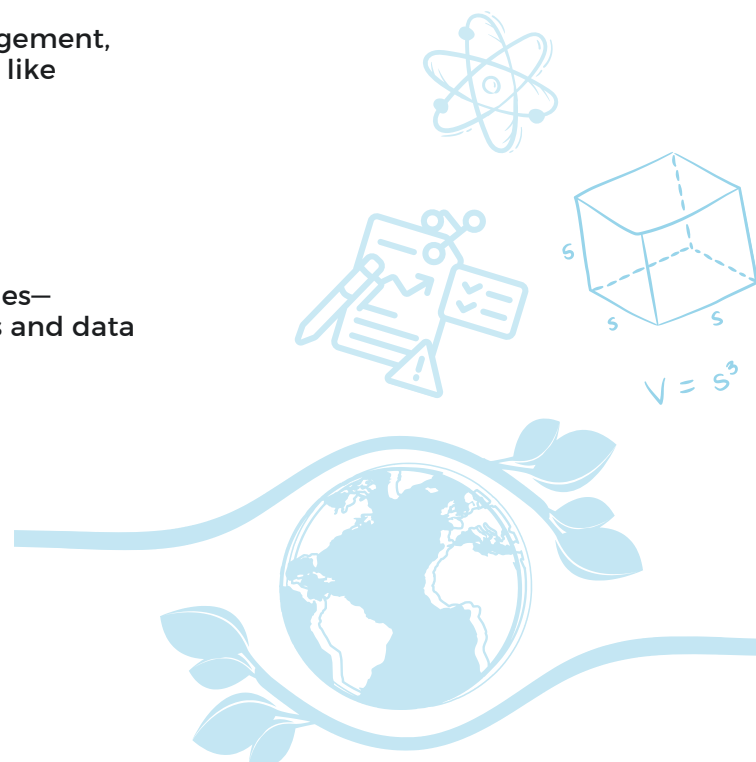
Organized by:

Fergusson College, Department of
[Geology/Physics/Mathematics]

✓ Who will be benefited?

Last year, undergraduate and postgraduate Students of:

- Geology
- Economics
- Mathematics and
- Industrial Mathematics with Computer Applications
- Physics
- Statistics
- Data Science
- Computer Science and Applications
- Environmental Science



Workshop Schedule Highlights

Day 1: Monday, 23rd June 2025

Tr Session	👤 Name	📄 Topic
11:00 - 12:00	Prof. Dr. Shyam Mude – Principal, Fergusson College (Autonomous), Pune:	Inaugural Talk
12:15 - 1:15	Prof. Dr. Duraiswami Raymond A, Dept. of Geology, SPPU	Introduction to Palaeontology / Geology
2:00 - 3:00	Dr. Hamsa Padmanabhan, Associate, The Green Concept former Principal Investigator, Swiss National Science Foundation Ambizione Grant	Lessons from the Earth's deadliest extinctions for today's climate crisis
3:00 onwards	All	Over tea discussion

Day 2: Tuesday, 24th June 2025

Tr Session	👤 Name	📄 Topic
11:00 - 12:00	Dr. Mukund Ramtirthkar, Assistant Professor, DESPU	Basics for Probability and Statistics
12:15 - 1:15	Dr. Girish Kulkarni, CEO and Cofounder, Vidhan.AI	Earth, Data, and Algorithms: Machine Learning Across Disciplines
2:00 - 3:00	Dr. Aboli Kulkarni, Co-Founder: The Green Concept Director: Green Concept Ecoscapes Pvt Ltd	From Equations to Ecosystems: The Power of Numbers in Ecology
3:00 onwards	All	Over tea discussion

Day 3: Wednesday, 25th June 2025

Tr Session	👤 Name	📄 Topic
11:00 - 12:00	Dr. Rohit Holkar, Assistant Professor at IISER Bhopal	Graph Theory and Applications to Geological Data
12:15 - 1:15	Mr. Lokender Singh Sohan Singh Rao, Mr. Adil Khan, Research Scholar, IISER Bhopal	Graph Theory and Applications to Geological Data
2:00 - 3:00	Dr. Argha Banerjee, Associate Professor, IISER Pune (Mode of Conduct - Online)	Geo-statistics
3:00 onwards	All	Over tea discussion

Day 4: Thursday, 26th June 2025

Tr Session	👤 Name	📄 Topic
11:00 - 12:00	Mr. Anay Shende, Geologist, Department of Land and Natural Resources (Mode of Conduct - Online)	Application of Geophysical Methods: Industrial and Regulatory Case Studies From the State of Hawaii (USA)
12:15 - 1:15	Dr. Ishita Ghoshal, Assistant Professor, Department of Economics, Fergusson College (Autonomous), Pune	Geo-Economics: A Glimpse into how the World Economy gets Shaped
2:00 - 3:00	Dr. Kumar Gaurav, Associate Professor, Department of Earth and Environmental Sciences, IISER Bhopal	Satellite Imaging
3:00 onwards	All	Over tea discussion

Day 5: Friday, 27th June 2025

Tr Session	👤 Name	📄 Topic
11:00 - 12:00	Mr. Hrishikesh Khaladkar, Assistant Professor, IMDR	Inverse theory and its applications to Geophysics
12:15 - 1:15	Dr. Nandini Kotharkar, Assistant Professor, Department of Biotechnology, Fergusson College(Autonomous), Pune	Lab to Incubator: The Way Ahead
2:15 - 3:00	Dr. Abhishek Iyer, Assistant Professor, Department of Physics, IIT Delhi	

